

# Hack the Hill Northwind Brief

2026-09-26

```
# Load Data
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ----- tidyverse 2.0.0 --
## v dplyr      1.2.1      v readr      2.2.0
## v forcats    1.0.1      v stringr   1.6.0
## v ggplot2    4.0.3      v tibble    3.3.1
## v lubridate  1.9.5      v tidyr     1.3.2
## v purrr      1.2.2
## -- Conflicts ----- tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()     masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(dplyr)
library(lubridate)
kpis <- read.csv("Northwind_Challenge_Data/northwind_monthly_kpis.csv")
complaints <- read.csv("Northwind_Challenge_Data/northwind_complaints.csv")
head(complaints)
```

```
##   complaint_id date_opened date_closed      status channel
## 1   NW-100001  2024-10-01  2024-10-30      Closed Web form
## 2   NW-100002  2024-10-01  2024-10-07      Closed Web form
## 3   NW-100003  2024-10-01  2024-10-16 Closed - reopened Phone
## 4   NW-100004  2024-10-01  2024-10-06      Closed Phone
## 5   NW-100005  2024-10-01  2024-10-03      Closed Phone
## 6   NW-100006  2024-10-01  2024-10-03      Closed Phone
##
##           category priority region source_system
## 1 Billing - disputed amount      P3 Ashford      SYS-01
## 2 Billing - disputed amount      P2 Ashford      SYS-03
## 3 Billing - disputed amount      P2 Ashford      SYS-01
## 4 Billing - disputed amount      P3 Fenwick      SYS-01
## 5 Billing - disputed amount      P2 Dunmoor      SYS-01
## 6 Service - poor communication      P3 Ashford      SYS-01
##
## transferred_between_systems sla_days days_to_close sla_breach reopened
## 1                        1      20      29      1      0
## 2                        0      10      6      0      0
## 3                        1      10      15      1      1
## 4                        1      20      5      0      0
## 5                        0      10      2      0      0
## 6                        0      20      2      0      0
##
##           resolution_action resolvable_by_information_only
## 1 Refund or credit applied      0
```

```
## 2 Bill corrected and re-issued 0
## 3 Bill corrected and re-issued 0
## 4 Refund or credit applied 0
## 5 Bill corrected and re-issued 0
## 6 Apology and manual process fix 0
## bill_correction_value account_id
## 1 73.57 ACC-943644
## 2 96.94 ACC-592536
## 3 71.61 ACC-830825
## 4 243.48 ACC-157354
## 5 180.64 ACC-372876
## 6 NA ACC-422289
```

```
staff <- read.csv("Additional CGI Files/northwind_contact_centre_staffing.csv")
meter <- read.csv("Northwind_Challenge_Data/northwind_meter_reads.csv")
```

```
# Number of open and reopened complaints
```

```
complaints_open = complaints |> filter(status == "Open") |> count(category, sort = TRUE) |> mutate(ratio = n/sum(n))
complaints_open
```

```
##           category    n    ratio
## 1 Billing - disputed amount 576 36.022514
## 2 Billing - estimated read 348 21.763602
## 3 Metering - no read taken 171 10.694184
## 4 Supply - interruption 125 7.817386
## 5 Service - poor communication 108 6.754221
## 6 Service - missed appointment 88 5.503440
## 7 Payment - plan or arrears 80 5.003127
## 8 Water - pressure or quality 60 3.752345
## 9 Other 43 2.689181
```

```
complaints_reopened = complaints |> filter(status == "Closed - reopened") |> count(category, sort = TRUE)
complaints_reopened
```

```
##           category    n    ratio
## 1 Billing - disputed amount 1205 31.912076
## 2 Billing - estimated read 723 19.147246
## 3 Metering - no read taken 458 12.129237
## 4 Supply - interruption 324 8.580508
## 5 Service - poor communication 288 7.627119
## 6 Service - missed appointment 247 6.541314
## 7 Payment - plan or arrears 221 5.852754
## 8 Water - pressure or quality 215 5.693856
## 9 Other 95 2.515890
```

```
# Total cost of certain complaints
```

```
resolution = complaints |> count(resolution_action, sort = TRUE) |> mutate(ratio = (n/sum(n))*100)
resolution
```

```
##           resolution_action    n    ratio
## 1 Bill corrected and re-issued 6089 23.957350
## 2 Information provided only 4783 18.818854
```

```
## 3          Meter visit required 4055 15.954517
## 4          Field repair required 2074  8.160214
## 5          Refund or credit applied 1785  7.023135
## 6                                     1599  6.291313
## 7          Payment plan amended 1122  4.414542
## 8          Appointment rebooked by agent 1087  4.276833
## 9 No action - explained to customer 1082  4.257161
## 10         Apology and manual process fix  976  3.840101
## 11         Compensation payment issued  764  3.005980
```

```
cost_resolution <- resolution |>
  mutate(total_cost = case_when(
    resolution_action == "Meter visit required" ~ n * 92,
    resolution_action == "Appointment rebooked by agent" ~ n * 40,
    resolution_action == "Bill corrected and re-issued" ~ n * 34,
    TRUE ~ 0
  ))
cost_resolution
```

```
##          resolution_action      n      ratio total_cost
## 1      Bill corrected and re-issued 6089 23.957350    207026
## 2      Information provided only 4783 18.818854         0
## 3      Meter visit required 4055 15.954517    373060
## 4      Field repair required 2074  8.160214         0
## 5      Refund or credit applied 1785  7.023135         0
## 6                                     1599  6.291313         0
## 7      Payment plan amended 1122  4.414542         0
## 8      Appointment rebooked by agent 1087  4.276833    43480
## 9 No action - explained to customer 1082  4.257161         0
## 10     Apology and manual process fix  976  3.840101         0
## 11     Compensation payment issued  764  3.005980         0
```

```
bill_correction = complaints |> filter(resolution_action == "Bill corrected and re-issued") |> summarize(
  bill_correction
```

```
##      total
## 1 1042472
```

```
refund = complaints |> filter(resolution_action == "Refund or credit applied") |> summarize(total = sum(
  refund
```

```
##      total
## 1 302717.4
```

```
transfers <- complaints |> count(transferred_between_systems) |> mutate(cost = case_when(
  transferred_between_systems == 1 ~ n * 121,
  transferred_between_systems == 0 ~ n * 68
))
transfers
```

```
##      transferred_between_systems      n      cost
## 1                                0 16546 1125128
## 2                                1  8870 1073270
```

```
resolution_ratios <- complaints |>
  count(transferred_between_systems, resolution_action) |>
  group_by(resolution_action) |>
  mutate(ratio = n / sum(n))
resolution_ratios
```

```
## # A tibble: 22 x 4
## # Groups:   resolution_action [11]
##   transferred_between_systems resolution_action      n ratio
##               <int> <chr>                <int> <dbl>
## 1               0 ""                        878 0.549
## 2               0 "Apology and manual process fix"      641 0.657
## 3               0 "Appointment rebooked by agent"      686 0.631
## 4               0 "Bill corrected and re-issued"    4083 0.671
## 5               0 "Compensation payment issued"     479 0.627
## 6               0 "Field repair required"      1368 0.660
## 7               0 "Information provided only"     3120 0.652
## 8               0 "Meter visit required"       2633 0.649
## 9               0 "No action - explained to customer"   730 0.675
## 10              0 "Payment plan amended"         740 0.660
## # i 12 more rows
```

```
region <- complaints |>
  count(transferred_between_systems, region) |>
  group_by(region) |>
  mutate(ratio = n / sum(n))
region
```

```
## # A tibble: 12 x 4
## # Groups:   region [6]
##   transferred_between_systems region      n ratio
##               <int> <chr>        <int> <dbl>
## 1               0 Ashford      2725 0.638
## 2               0 Barrowdale    2753 0.641
## 3               0 Calderfield  2705 0.657
## 4               0 Dunmoor      2836 0.656
## 5               0 Eastmarch    2747 0.657
## 6               0 Fenwick      2780 0.657
## 7               1 Ashford      1544 0.362
## 8               1 Barrowdale    1542 0.359
## 9               1 Calderfield  1413 0.343
## 10              1 Dunmoor      1484 0.344
## 11              1 Eastmarch    1433 0.343
## 12              1 Fenwick      1454 0.343
```

```
# Increased Rate of Meter Related Complaints in Barrowdale and Dunmoor
condensed_region <- complaints |>
  mutate(region_group = case_when(
    region %in% c("Barrowdale", "Dunmoor") ~ "Barrowdale & Dunmoor",
    TRUE ~ "Other Regions"
  ))
condensed_region |> count(region_group)
```

```
##           region_group      n
## 1 Barrowdale & Dunmoor  8615
## 2           Other Regions 16801
```

```
billing_rate <- condensed_region |>
  count(category, region_group) |>
  group_by(region_group) |>
  mutate(ratio = n / sum(n))
billing_rate
```

```
## # A tibble: 18 x 4
## # Groups:   region_group [2]
##   category                region_group      n ratio
##   <chr>                  <chr>      <int> <dbl>
## 1 Billing - disputed amount Barrowdale & Dunmoor  2782 0.323
## 2 Billing - disputed amount Other Regions        5278 0.314
## 3 Billing - estimated read  Barrowdale & Dunmoor  2045 0.237
## 4 Billing - estimated read  Other Regions        2788 0.166
## 5 Metering - no read taken Barrowdale & Dunmoor  1633 0.190
## 6 Metering - no read taken Other Regions        1487 0.0885
## 7 Other                   Barrowdale & Dunmoor   132 0.0153
## 8 Other                   Other Regions         539 0.0321
## 9 Payment - plan or arrears Barrowdale & Dunmoor   383 0.0445
## 10 Payment - plan or arrears Other Regions        1218 0.0725
## 11 Service - missed appointment Barrowdale & Dunmoor  403 0.0468
## 12 Service - missed appointment Other Regions        1315 0.0783
## 13 Service - poor communication Barrowdale & Dunmoor  438 0.0508
## 14 Service - poor communication Other Regions        1470 0.0875
## 15 Supply - interruption Barrowdale & Dunmoor   506 0.0587
## 16 Supply - interruption Other Regions        1686 0.100
## 17 Water - pressure or quality Barrowdale & Dunmoor   293 0.0340
## 18 Water - pressure or quality Other Regions        1020 0.0607
```

```
write.table(billing_rate, "clipboard", sep = "\t", row.names = FALSE, quote = FALSE)
```

```
# Number of complaints by priority
open <- complaints |> filter(status == "Open")
priority_count <- open |> count(priority)
priority_count
```

```
##   priority      n
## 1      P1      35
## 2      P2     219
## 3      P3    1345
```

```
p1 <- open |> filter(priority == "P1") |> arrange(region)
p1
```

```
##   complaint_id date_opened date_closed status  channel
## 1     NW-124627  2026-09-14           Open   Phone
## 2     NW-124848  2026-09-17           Open   Phone
```

|       |                              |            |             |               |
|-------|------------------------------|------------|-------------|---------------|
| ## 3  | NW-125001                    | 2026-09-21 | Open        | Phone         |
| ## 4  | NW-125134                    | 2026-09-24 | Open        | Social        |
| ## 5  | NW-125273                    | 2026-09-28 | Open        | Web form      |
| ## 6  | NW-125340                    | 2026-09-29 | Open        | Phone         |
| ## 7  | NW-125372                    | 2026-09-30 | Open        | Phone         |
| ## 8  | NW-124742                    | 2026-09-15 | Open        | Phone         |
| ## 9  | NW-124946                    | 2026-09-21 | Open        | Phone         |
| ## 10 | NW-125116                    | 2026-09-24 | Open        | Web form      |
| ## 11 | NW-125269                    | 2026-09-28 | Open        | Email         |
| ## 12 | NW-125403                    | 2026-09-30 | Open        | Phone         |
| ## 13 | NW-125413                    | 2026-09-30 | Open        | Phone         |
| ## 14 | NW-124795                    | 2026-09-16 | Open        | Phone         |
| ## 15 | NW-124904                    | 2026-09-18 | Open        | Phone         |
| ## 16 | NW-125098                    | 2026-09-23 | Open        | Phone         |
| ## 17 | NW-125177                    | 2026-09-25 | Open        | Social        |
| ## 18 | NW-125258                    | 2026-09-28 | Open        | Phone         |
| ## 19 | NW-125342                    | 2026-09-29 | Open        | Web form      |
| ## 20 | NW-124897                    | 2026-09-18 | Open        | Phone         |
| ## 21 | NW-124978                    | 2026-09-21 | Open        | Phone         |
| ## 22 | NW-125102                    | 2026-09-24 | Open        | Phone         |
| ## 23 | NW-125324                    | 2026-09-29 | Open        | Email         |
| ## 24 | NW-125398                    | 2026-09-30 | Open        | Phone         |
| ## 25 | NW-124358                    | 2026-09-07 | Open        | Phone         |
| ## 26 | NW-124903                    | 2026-09-18 | Open        | Phone         |
| ## 27 | NW-125106                    | 2026-09-24 | Open        | Social        |
| ## 28 | NW-125124                    | 2026-09-24 | Open        | Email         |
| ## 29 | NW-124487                    | 2026-09-09 | Open        | Phone         |
| ## 30 | NW-125037                    | 2026-09-22 | Open        | Web form      |
| ## 31 | NW-125050                    | 2026-09-22 | Open        | Phone         |
| ## 32 | NW-125166                    | 2026-09-25 | Open        | Email         |
| ## 33 | NW-125285                    | 2026-09-28 | Open        | Web form      |
| ## 34 | NW-125287                    | 2026-09-28 | Open        | Phone         |
| ## 35 | NW-125309                    | 2026-09-29 | Open        | Phone         |
| ##    | category                     | priority   | region      | source_system |
| ## 1  | Service - missed appointment | P1         | Ashford     | SYS-03        |
| ## 2  | Service - poor communication | P1         | Ashford     | SYS-05        |
| ## 3  | Billing - disputed amount    | P1         | Ashford     | SYS-04        |
| ## 4  | Billing - disputed amount    | P1         | Ashford     | SYS-01        |
| ## 5  | Payment - plan or arrears    | P1         | Ashford     | SYS-03        |
| ## 6  | Billing - disputed amount    | P1         | Ashford     | SYS-03        |
| ## 7  | Water - pressure or quality  | P1         | Ashford     | SYS-03        |
| ## 8  | Billing - disputed amount    | P1         | Barrowdale  | SYS-01        |
| ## 9  | Billing - disputed amount    | P1         | Barrowdale  | SYS-05        |
| ## 10 | Billing - estimated read     | P1         | Barrowdale  | SYS-05        |
| ## 11 | Metering - no read taken     | P1         | Barrowdale  | SYS-05        |
| ## 12 | Billing - disputed amount    | P1         | Barrowdale  | SYS-01        |
| ## 13 | Billing - disputed amount    | P1         | Barrowdale  | SYS-01        |
| ## 14 | Billing - estimated read     | P1         | Calderfield | SYS-05        |
| ## 15 | Billing - disputed amount    | P1         | Calderfield | SYS-05        |
| ## 16 | Supply - interruption        | P1         | Calderfield | SYS-04        |
| ## 17 | Billing - disputed amount    | P1         | Calderfield | SYS-04        |
| ## 18 | Service - missed appointment | P1         | Calderfield | SYS-03        |
| ## 19 | Supply - interruption        | P1         | Calderfield | SYS-05        |
| ## 20 | Billing - estimated read     | P1         | Dunmoor     | SYS-05        |

|       |                              |                                |                       |            |          |    |
|-------|------------------------------|--------------------------------|-----------------------|------------|----------|----|
| ## 21 | Billing - disputed amount    | P1                             | Dunmoor               | SYS-04     |          |    |
| ## 22 | Metering - no read taken     | P1                             | Dunmoor               | SYS-04     |          |    |
| ## 23 | Metering - no read taken     | P1                             | Dunmoor               | SYS-04     |          |    |
| ## 24 | Service - missed appointment | P1                             | Dunmoor               | SYS-05     |          |    |
| ## 25 | Billing - estimated read     | P1                             | Eastmarch             | SYS-01     |          |    |
| ## 26 | Billing - disputed amount    | P1                             | Eastmarch             | SYS-01     |          |    |
| ## 27 | Billing - disputed amount    | P1                             | Eastmarch             | SYS-01     |          |    |
| ## 28 | Billing - disputed amount    | P1                             | Eastmarch             | SYS-03     |          |    |
| ## 29 | Supply - interruption        | P1                             | Fenwick               | SYS-03     |          |    |
| ## 30 | Other                        | P1                             | Fenwick               | SYS-01     |          |    |
| ## 31 | Metering - no read taken     | P1                             | Fenwick               | SYS-04     |          |    |
| ## 32 | Billing - estimated read     | P1                             | Fenwick               | SYS-01     |          |    |
| ## 33 | Billing - disputed amount    | P1                             | Fenwick               | SYS-05     |          |    |
| ## 34 | Supply - interruption        | P1                             | Fenwick               | SYS-04     |          |    |
| ## 35 | Billing - disputed amount    | P1                             | Fenwick               | SYS-03     |          |    |
| ##    | transferred_between_systems  | sla_days                       | days_to_close         | sla_breach | reopened |    |
| ## 1  |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 2  |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 3  |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 4  |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 5  |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 6  |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 7  |                              | 0                              | 5                     | NA         | 0        | 0  |
| ## 8  |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 9  |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 10 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 11 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 12 |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 13 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 14 |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 15 |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 16 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 17 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 18 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 19 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 20 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 21 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 22 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 23 |                              | 0                              | 5                     | NA         | 0        | 0  |
| ## 24 |                              | 0                              | 5                     | NA         | 0        | 0  |
| ## 25 |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 26 |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 27 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 28 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 29 |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 30 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 31 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 32 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 33 |                              | 1                              | 5                     | NA         | 1        | 0  |
| ## 34 |                              | 0                              | 5                     | NA         | 1        | 0  |
| ## 35 |                              | 1                              | 5                     | NA         | 1        | 0  |
| ##    | resolution_action            | resolvable_by_information_only | bill_correction_value |            |          |    |
| ## 1  |                              |                                | NA                    |            |          | NA |
| ## 2  |                              |                                | NA                    |            |          | NA |

|                  |    |    |
|------------------|----|----|
| ## 3             | NA | NA |
| ## 4             | NA | NA |
| ## 5             | NA | NA |
| ## 6             | NA | NA |
| ## 7             | NA | NA |
| ## 8             | NA | NA |
| ## 9             | NA | NA |
| ## 10            | NA | NA |
| ## 11            | NA | NA |
| ## 12            | NA | NA |
| ## 13            | NA | NA |
| ## 14            | NA | NA |
| ## 15            | NA | NA |
| ## 16            | NA | NA |
| ## 17            | NA | NA |
| ## 18            | NA | NA |
| ## 19            | NA | NA |
| ## 20            | NA | NA |
| ## 21            | NA | NA |
| ## 22            | NA | NA |
| ## 23            | NA | NA |
| ## 24            | NA | NA |
| ## 25            | NA | NA |
| ## 26            | NA | NA |
| ## 27            | NA | NA |
| ## 28            | NA | NA |
| ## 29            | NA | NA |
| ## 30            | NA | NA |
| ## 31            | NA | NA |
| ## 32            | NA | NA |
| ## 33            | NA | NA |
| ## 34            | NA | NA |
| ## 35            | NA | NA |
| ## account_id    |    |    |
| ## 1 ACC-928088  |    |    |
| ## 2 ACC-527078  |    |    |
| ## 3 ACC-842092  |    |    |
| ## 4 ACC-808598  |    |    |
| ## 5 ACC-870326  |    |    |
| ## 6 ACC-582210  |    |    |
| ## 7 ACC-461418  |    |    |
| ## 8 ACC-506765  |    |    |
| ## 9 ACC-686903  |    |    |
| ## 10 ACC-193667 |    |    |
| ## 11 ACC-166686 |    |    |
| ## 12 ACC-918115 |    |    |
| ## 13 ACC-189137 |    |    |
| ## 14 ACC-217185 |    |    |
| ## 15 ACC-556054 |    |    |
| ## 16 ACC-202143 |    |    |
| ## 17 ACC-533428 |    |    |
| ## 18 ACC-584013 |    |    |
| ## 19 ACC-160484 |    |    |
| ## 20 ACC-792199 |    |    |

```
## 21 ACC-565104
## 22 ACC-335815
## 23 ACC-358826
## 24 ACC-727270
## 25 ACC-687033
## 26 ACC-878405
## 27 ACC-551330
## 28 ACC-365650
## 29 ACC-252733
## 30 ACC-980210
## 31 ACC-296387
## 32 ACC-914967
## 33 ACC-216483
## 34 ACC-600452
## 35 ACC-618524
```

```
# Staff in each area
```

```
condensed_staff <- staff |>
  mutate(region_group = case_when(
    region %in% c("Barrowdale", "Dunmoor") ~ "Barrowdale & Dunmoor",
    TRUE ~ "Other Regions"))
head(condensed_staff)
```

```
##      month      region agent_fte open_vacancies attrition_rate_12m
## 1 2024-10    Ashford         84             4           0.131
## 2 2024-10 Barrowdale         66             3           0.144
## 3 2024-10 Calderfield         73             4           0.117
## 4 2024-10    Dunmoor         51             2           0.158
## 5 2024-10 Eastmarch         62             3           0.130
## 6 2024-10    Fenwick         44             2           0.122
##      complaints_opened_per_agent note      region_group
## 1                      2.1      Other Regions
## 2                      2.4 Barrowdale & Dunmoor
## 3                      2.2      Other Regions
## 4                      2.8 Barrowdale & Dunmoor
## 5                      2.1      Other Regions
## 6                      3.3      Other Regions
```

```
# Regulator Metrics
```

```
closed <- complaints |> filter(status == "Closed")
on_time_closure <- closed |> count(status, sla_breach) |>
  group_by(status) |>
  mutate(ratio = n/sum(n))
on_time_closure
```

```
## # A tibble: 2 x 4
## # Groups:   status [1]
##   status sla_breach    n ratio
##   <chr>      <int> <int> <dbl>
## 1 Closed         0  5148 0.257
## 2 Closed         1 14893 0.743
```

```

# Combined monthly data, predicting regulator satisfaction from data we have
monthly_sla_summary <- complaints |>
mutate(
  # 1. Convert Excel serial numbers to dates
  date_opened = as.Date(date_opened, origin = "1899-12-30"),
  date_closed = as.Date(date_closed, origin = "1899-12-30"),

  # 2. Create a clean month column (snapped to the 1st of each month)
  month_opened = floor_date(date_opened, "month"),

  # 3. Calculate duration and check target
  duration_days = as.numeric(difftime(date_closed, date_opened, units = "days")),
  met_target = case_when(
    priority == "P1" ~ duration_days <= 5,
    priority == "P2" ~ duration_days <= 10,
    priority == "P3" ~ duration_days <= 20,
    TRUE ~ FALSE
  )
) |>
# 4. Group by month to calculate metrics per period
group_by(month_opened) |>
summarise(
  total_complaints = n(),
  closed_on_time = sum(met_target, na.rm = TRUE),
  pct_within_target = (closed_on_time / total_complaints) * 100
) |>
arrange(month_opened)

monthly_sla_summary

```

```

## # A tibble: 24 x 4
##   month_opened total_complaints closed_on_time pct_within_target
##   <date>          <int>          <int>          <dbl>
## 1 2024-10-01           912            511            56.0
## 2 2024-11-01           830            416            50.1
## 3 2024-12-01           951            465            48.9
## 4 2025-01-01           985            443            45.0
## 5 2025-02-01           851            353            41.5
## 6 2025-03-01           901            350            38.8
## 7 2025-04-01          1023            400            39.1
## 8 2025-05-01          1004            350            34.9
## 9 2025-06-01           982            323            32.9
## 10 2025-07-01          1222            338            27.7
## # i 14 more rows

```

```

monthly_reopened_summary <- complaints |>
mutate(
  # 1. Convert Excel serial numbers to proper dates
  date_opened = as.Date(date_opened, origin = "1899-12-30"),
  date_closed = as.Date(date_closed, origin = "1899-12-30"),

  # 2. Create a clean month column (snapped to the 1st of each month)
  month_opened = floor_date(date_opened, "month"),

```

```

# 3. Check if it was reopened and closed within 3 months of opening
reopened_in_3_months = (status == "Closed - reopened") & (date_closed <= date_opened + months(3))
) |>
# 4. Group by month to calculate the metrics per period
group_by(month_opened) |>
summarise(
  total_complaints = n(),
  reopened_on_time = sum(reopened_in_3_months, na.rm = TRUE),
  pct_reopened_in_3_months = (reopened_on_time / total_complaints) * 100
) |>
arrange(month_opened)

monthly_reopened_summary

```

```

## # A tibble: 24 x 4
##   month_opened total_complaints reopened_on_time pct_reopened_in_3_months
##   <date>          <int>          <int>          <dbl>
## 1 2024-10-01           912            157            17.2
## 2 2024-11-01           830            130            15.7
## 3 2024-12-01           951            140            14.7
## 4 2025-01-01           985            155            15.7
## 5 2025-02-01           851            142            16.7
## 6 2025-03-01           901            147            16.3
## 7 2025-04-01          1023            163            15.9
## 8 2025-05-01          1004            167            16.6
## 9 2025-06-01           982            152            15.5
## 10 2025-07-01          1222            208            17.0
## # i 14 more rows

```

```

model <- lm(kpis$regulator_satisfaction_score_of_5 ~ monthly_sla_summary$pct_within_target + monthly_reopened_summary$pct_reopened_in_3_months)
summary(model)

```

```

##
## Call:
## lm(formula = kpis$regulator_satisfaction_score_of_5 ~ monthly_sla_summary$pct_within_target +
##   monthly_reopened_summary$pct_reopened_in_3_months)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.14286 -0.03899  0.00734  0.04880  0.12285
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)    2.410926   0.066285  36.372   < 2e-16 ***
## monthly_sla_summary$pct_within_target  0.029932   0.001057  28.319   < 2e-16 ***
## monthly_reopened_summary$pct_reopened_in_3_months  0.019654   0.004918   3.997 0.000655 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
##  
## Residual standard error: 0.07321 on 21 degrees of freedom  
## Multiple R-squared:  0.9826, Adjusted R-squared:  0.9809  
## F-statistic: 591.9 on 2 and 21 DF,  p-value: < 2.2e-16
```